

Carla J. Becker

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Full resume available at:



SKILLS & PROFICIENCIES

Program management, early-stage project ramping, product engineering, process development — Machine-learning, optimization, numerical methods, controls engineering — Bring-up, operation, and maintenance of manufacturing machinery, analytical instrumentation, and electronic equipment — Python, MATLAB, C, LaTeX, Javascript, Unix, Bash

RESEARCH & INDUSTRY EXPERIENCE

- **Senior Product Engineer**, Sakuu Corporation *May 2018 - May 2022*
 - **Program Management** — Single-handedly managed, envisioned, and guided development of a browser-based integrated data management, machine learning, and advanced battery analytics system. Project had a \$300,000 budget and lasted 1.5 years. — Drafted statements of work and consulting agreements. Developed all company safety procedures and SOPs.
 - **Product Engineering** — Experience with powder milling, coin/pouch cell assembly, tape casting, slurry preparation, battery assembly and test, and powder bed 3D printing. — Design of experiment, process development, contract manufacturing. — Point-of-contact for fixing/repairing malfunctioning analytical instrumentation.
 - **People Management** — Daily managed a 10-person team of technicians and associate engineers on the R&D lab floor. — Managed 4 different interns to complete their own independent projects.
 - **Early-stage Project Ramping** — Developed strategy to grow a battery modeling and simulation business unit. Interviewed potential candidates and evaluated commercial finite element software. Defined 5-year scope, objectives, and success metrics. — Onboarded and trained all non-executive, technical staff on techniques, procedures, machinery, instrumentation, and theory until the startup headcount reached 50.
- **Ph.D. Intern, Design for Manufacturing**, Intel Corporation *February 2025 - August 2025*
 - Developed a web app for automated gathering, visualization, manipulation, and live editing of component characteristic information. Experience with OAuth, Snowflake, Flask, Docker, Kubernetes, email automation, and remote FEA simulation.
 - Evaluated 18 machine learning methods for predicting solder joint crack severity from board layout and stackup.
 - Independently ideated a novel method for running high-fidelity system-level FEA simulations in a fraction of the standard computation time by leveraging neural nets and global-local simulation.
- **Intern, Business Operations Management**, Intel Corporation *June 2022 - May 2023*
 - Streamlined hiring and finance operations within the Developer Software Engineering group by automating
 - 1) headcount reporting, 2) tracking of organization wide OKRs from JIRA data, and 3) generation of new job requisitions using Python Selenium, the PowerBI Javascript API, and Microsoft Power Automate.
- **Graduate Student Researcher**, University of California Berkeley *May 2022 - Present*
 - **Browser App for Composites Discovery, The Materials Project** — Built an app to recommend composite material components based on user-defined properties for an ultimate material. Public GitHub repo [here](#). — Developed an object-oriented framework in Python which uses, and will be integrated into, the Materials Project API. — Tested a suite of optimization techniques and developed faster versions in C and Cython.
 - **Fellow at the AI Institute for Next-Generation Food Systems** — Developed an extensible coupled, nonlinear ODE model for growth of a general crop. Public GitHub repo [here](#). — Developed a ray tracing solver for a digital twin of an indoor farm, optimizing LED configuration for maximum light absorption. Resulted in [publication](#). Public GitHub repo [here](#).
- **Student Assistant, College of Engineering**, University of California Berkeley *May 2022 - Present*
 - **Outstanding Graduate Student Instructor 2024**, Modeling and Simulation of Advanced Manufacturing Processes.
 - **Backend Software Development, Capstone Connect** — Automated optimal matching based on mutual ranking between industry sponsors and Masters of Engineering students, who must complete a capstone project to graduate, and automated all document production and email communication. — All work done in Google AppScript (Javascript).
 - **Head Student Staff Assistant, Dubai Electricity and Water Authority (DEWA)** — Administered the engineering certificate program for students at DEWA through the UC Berkeley Extension. Organized student enrollment, grading, forums, and course resources. — Managed many stakeholders: the program manager and program director from DEWA, the UC Berkeley registrar, staff from the UC Berkeley Extension office, professors, and teaching assistants.

EDUCATION

- **University of California Berkeley**
Ph.D., Mechanical Engineering; Expected: May 2026
M.S., Materials Science & Engineering; Earned: May 2022
- **Harvey Mudd College**
B.S. Physics, B.S. Chemistry; Earned: May 2018

RELEVANT & PLANNED GRADUATE COURSEWORK

Finite Element Method · Numerical Solutions of ODEs/PDEs · State Estimation and Kalman Filtering · Model Predictive Control · Data Science for Energy · Materials Science of Energy Conversion and Storage · Power Electronics · Semiconductor Manufacturing - Bring-up (Spring 2025) · Semiconductor Manufacturing - Tape-out (Fall 2025)